

Phishing Email Investigation Report

Brazilian Tax Scam (IRPF) — Malicious PDF Attachment Analysis | Phishing Pot Lab

Incident ID	INC-2026-002	Analyst	Gaurav
Date	2026-06-23	Severity	HIGH
Sample	sample-1000.eml (Phishing Pot)	Status	Confirmed — True Positive
MITRE ATT&CK	T1566.001 — Spearphishing Attachment	Tool	VirusTotal, MXToolbox, AbuseIPDB

1. Executive Summary

A phishing email impersonating the **Brazilian Federal Revenue Service (IRPF)** was analyzed from the Phishing Pot public dataset (sample-1000.eml). The email was sent from a throwaway Gmail account (**prestonconstance587@gmail.com**) using Gmail's SMTP infrastructure to evade email security filters. The email contained a **malicious PDF attachment (csWuYjyqO2IR.pdf)** classified as **Trojan:PDF/Phishing.A** by 23 out of 63 antivirus engines on VirusTotal. This is a **True Positive — HIGH severity** phishing incident using a tax refund lure to trick victims into opening a malicious attachment.

2. Email Header Analysis

Field	Value	Verdict
From	prestonconstance587@gmail.com	■ Throwaway Gmail account
Return-Path	prestonconstance587@gmail.com	■■ Matches From — no spoofing but fake account
Subject	Liberacao de IRPF	■ Brazilian tax refund lure (IRPF)
Sending IP	209.85.160.178	■■ Google LLC (Gmail SMTP) — legitimate service abused
Origin IP	20.97.213.223	■ Microsoft Azure VM — attacker-controlled
SPF	PASS	■■ Passes — Gmail is authorized sender
DKIM	PASS	■■ Passes — Gmail signed the email
DMARC	PASS	■■ Passes — Attacker used real Gmail account
Date	Wed, 26 Jul 2023 17:59:01 +0000	■■ Historical sample from Phishing Pot dataset
Attachment	csWuYjyqO2IR.pdf	■ Random filename — highly suspicious

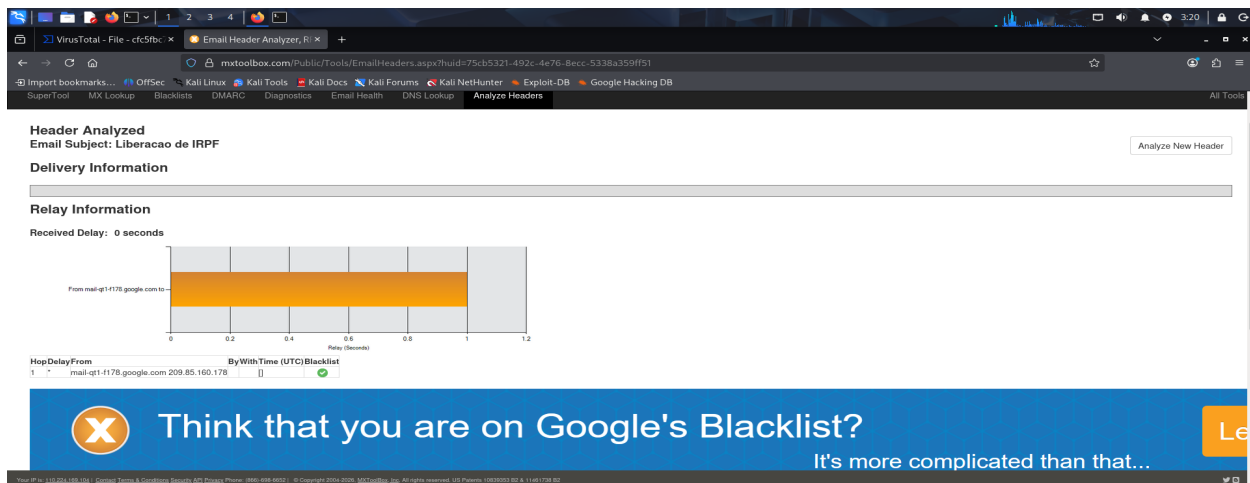


Figure 1: MXToolbox Email Header Analysis — relay path from Gmail to Outlook

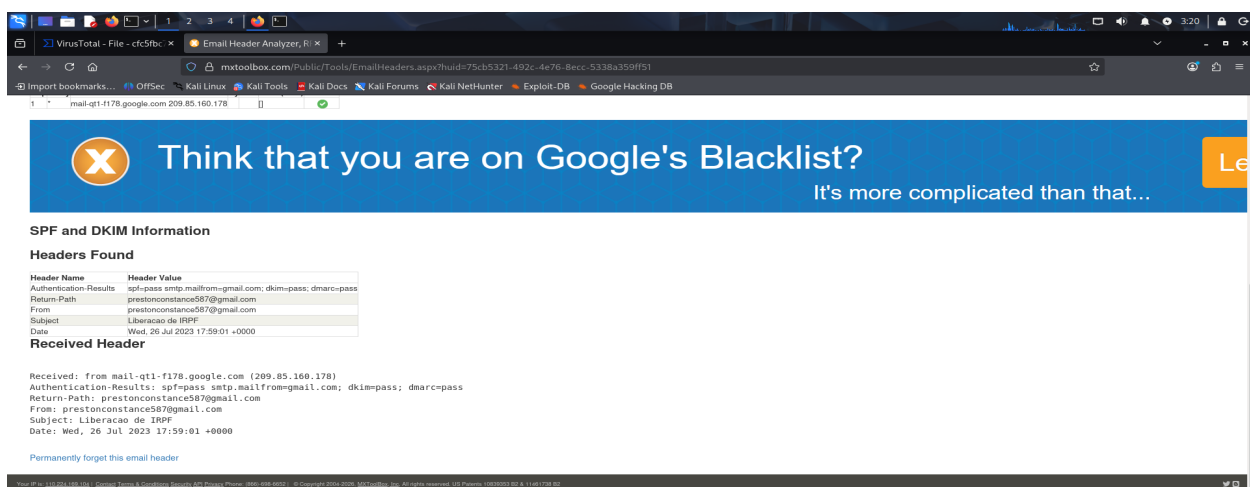


Figure 2: MXToolbox SPF/DKIM/DMARC results — all PASS (Gmail legitimate service abuse)

3. Key Finding — Legitimate Service Abuse

All three email authentication checks (SPF, DKIM, DMARC) **PASS** — this does NOT mean the email is safe. The attacker deliberately used a **real Gmail account** to send the phishing email, knowing that Gmail's infrastructure would pass all authentication checks. This is a common attacker tactic called '**Legitimate Service Abuse**' — using trusted platforms (Gmail, OneDrive, Google Docs) to bypass email security filters that rely on SPF/DKIM/DMARC validation.

Tactic	Detail
Attacker Used	Free Gmail account (prestonconstance587@gmail.com)
Why Gmail?	Gmail passes SPF/DKIM/DMARC — bypasses most email filters
Real Origin	Microsoft Azure VM (20.97.213.223) — attacker-controlled server
Lure Used	Brazilian tax refund (IRPF) — urgency + financial gain
Language	Portuguese — targeted Brazilian victims

4. Malicious Attachment Analysis

The PDF attachment was safely extracted using Python and analyzed via VirusTotal:

Property	Value
Filename	csWuYjyqO2IR.pdf (random string — obfuscation tactic)
File Size	97,566 bytes (95.28 KB)
MD5 Hash	f1b1ac7839d2b2a1e458b5d87c7b2ca2
SHA256 Hash	cfc5fbc759dcc599c8329dd94f9364394c7b1e875d6ff515c7337e00fb1f30cf
VirusTotal	23/63 security vendors flagged as MALICIOUS
Threat Label	trojan.abphisher/atmn
Threat Category	Trojan Phishing
Family Labels	abphisher, atmn, ocnw
AliCloud	Trojan:PDF/Phishing.A
ALYac	Trojan.GenericKD.72313796
Arcabit	Trojan.Generic.D44F6BC4

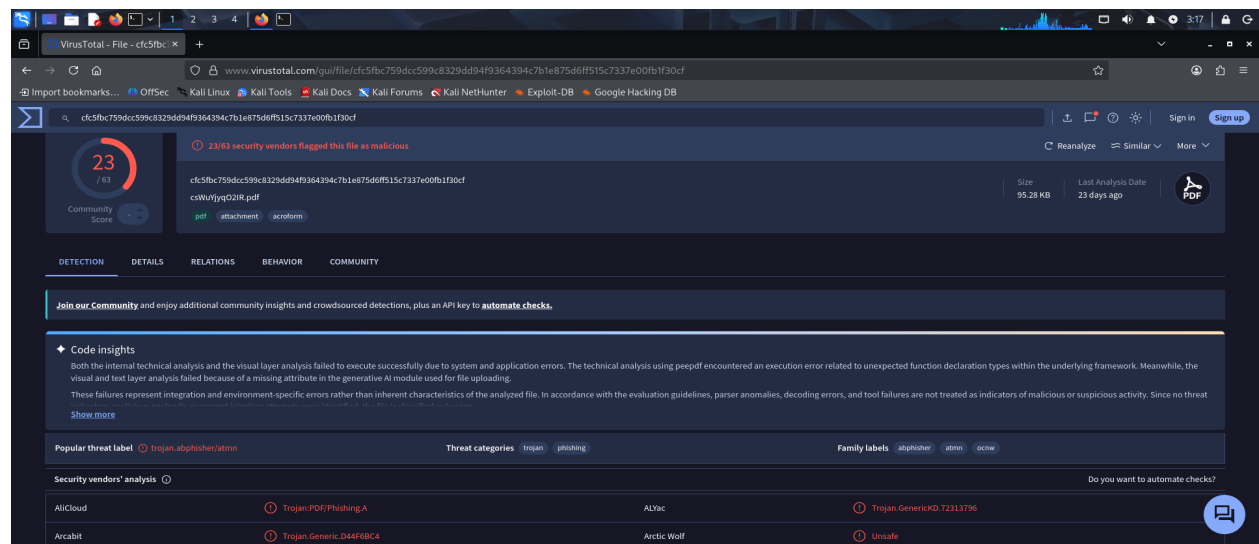


Figure 3: VirusTotal result — 23/63 vendors flagged PDF as malicious Trojan (Phishing)

5. Indicator of Compromise (IOC) Analysis

IOC Type	Value	Tool	Result
Sender Email	prestonconstance587@gmail.com	Manual	■ Throwaway Gmail — attacker created
Sending IP	209.85.160.178	AbuseIPDB	■■ Google LLC — 84 reports, 0% abuse (legitimate Gmail server)
Origin IP	20.97.213.223	Manual	■ Microsoft Azure VM — attacker-controlled infrastructure
PDF Hash (SHA256)	cfc5fbc759...f30cf	VirusTotal	■ 23/63 MALICIOUS — Trojan:PDF/Phishing.A
Subject	Liberacao de IRPF	Manual	■ Tax refund lure — social engineering
Attachment Name	csWuYjyqO2IR.pdf	Manual	■ Random filename — obfuscation tactic

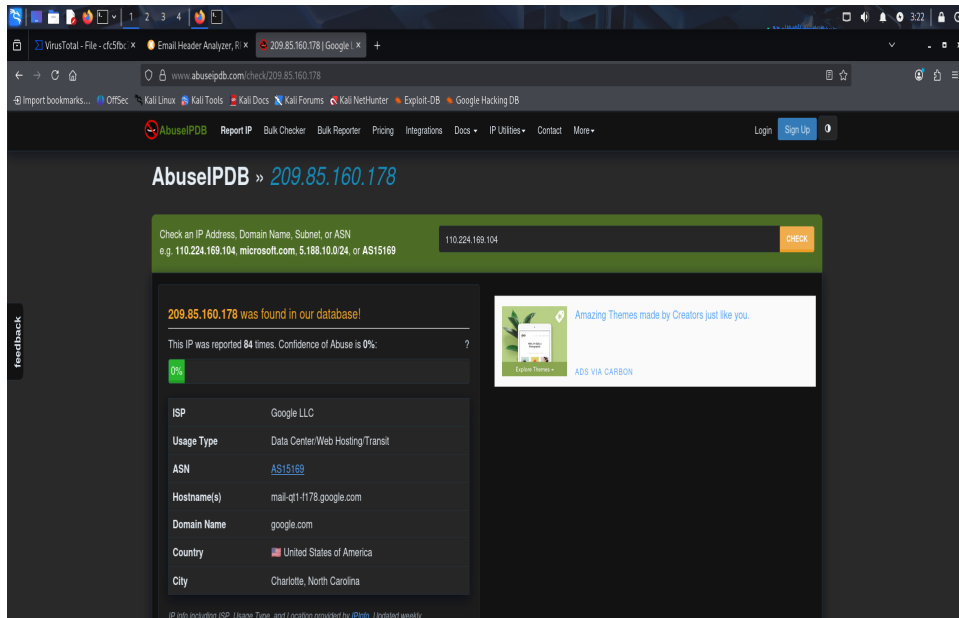


Figure 4: AbuseIPDB — 209.85.160.178 (Google Gmail SMTP), 84 reports, 0% abuse confidence

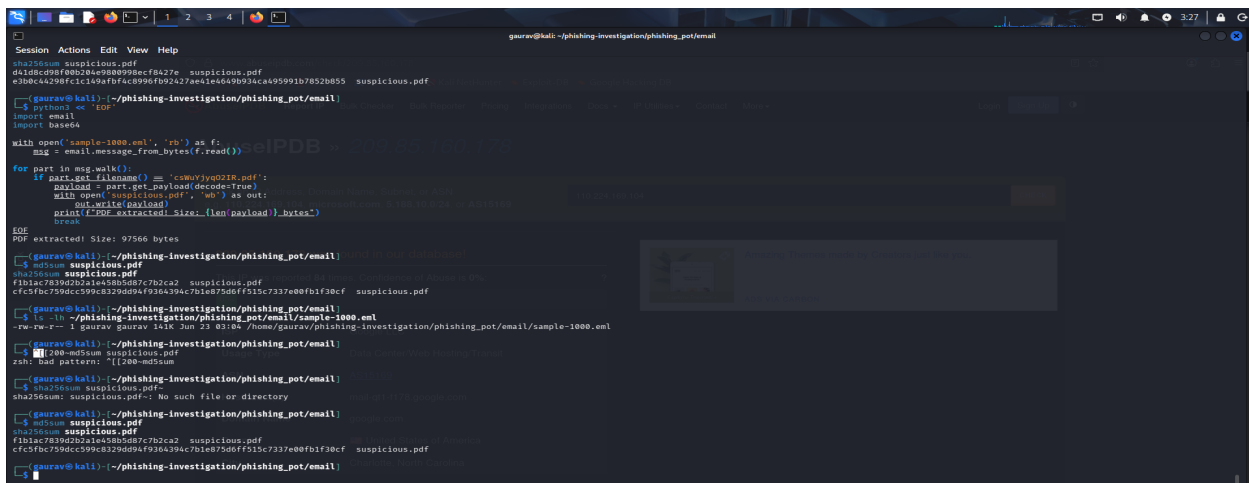


Figure 5: Terminal — PDF extraction using Python (97566 bytes) and SHA256/MD5 hash generation

6. MITRE ATT&CK; Mapping

Tactic	Technique	ID
Initial Access	Phishing: Spearphishing Attachment	T1566.001
Execution	User Execution: Malicious File	T1204.002
Defense Evasion	Masquerading (random PDF filename)	T1036
Defense Evasion	Abuse of Legitimate Service (Gmail)	T1550
Credential Access	Phishing for Information	T1598

7. Verdict & Classification

Classification	True Positive — MALICIOUS PHISHING EMAIL
Attack Type	Phishing with Malicious PDF Attachment (Trojan:PDF/Phishing.A)

Target	Brazilian users — Tax refund (IRPF) social engineering lure
Severity	HIGH — Malicious payload confirmed by 23/63 AV engines
Confidence	HIGH — Multiple IOCs confirmed across VirusTotal, MXToolbox, AbuseIPDB

8. Recommended Remediation Actions

Priority	Action	Owner
IMMEDIATE	Block sender domain/email: prestonconstance587@gmail.com	Email Security Team
IMMEDIATE	Quarantine/purge email from all affected mailboxes	Email Admin
IMMEDIATE	Block PDF hash on endpoint security (EDR/AV)	SOC / IT
HIGH	Block origin IP 20.97.213.223 at perimeter firewall	Network Team
HIGH	Alert users about IRPF/tax refund phishing campaign	Security Awareness
HIGH	Add IOCs (hash, email, IP) to SIEM detection rules	SOC Engineer
MEDIUM	Check if any users opened the PDF attachment — review EDR logs	L2 Analyst
MEDIUM	Submit PDF hash to threat intel sharing platforms (MISP)	Threat Intel
LOW	Update email security rules to flag Portuguese-language emails with PDF attachments	Email Security
